

Industrial Policy in China: Organization, Development and Methods.

(Note: My writing sample is a small part of a 25-page research policy paper commissioned by the Eteron Institution)

In March 2024, Chinese company Xiaomi announced the production of electric cars using AI ecosystems that connect devices in all areas of everyday life. The "Human x Car x Home" ecosystem, as reflected in the Xiaomi SU7, managed to surpass multinationals with many years of experience in the field, such as Tesla and Ford, in sales within eight months. During the same period, an article on the cover of the English print edition of China Daily was titled "Why didn't ChatGPT first appear in China?" A few months later, DeepSeek made its appearance using comparatively cheaper and faster means.

So, how did a mobile phone company manage to surpass major car manufacturers like Tesla and Ford in such a short period of time? What are the reasons, policies, and institutions behind the changes in Chinese industrial policy? What should the decision-making processes be when technological developments are moving faster than society and politics can assimilate? The answers to these questions will be provided through an in-depth analysis of the course taken by industrial policy and the institutional-administrative model of modern China.

Administrative and Economic Model of Modern China.

China's contemporary industrial policy reflects an interwoven legacy of historical imperialistic memory, responses to multiple geopolitical crises, and prioritized objectives of security (ānquán, 安全) and stability (wéiwěn, 维稳). While the trajectory of its industrial strategy has been consistently upward, a coherent framework and organized institutional structure started to take shape only in recent years.

During the 1960s and 1970s, East Asian countries established a development system characterized by state intervention in industry, a strong bureaucracy, and close cooperation with large enterprises (Johnson, 1982). The "developmental state," as it was termed, was regarded as a model for crafting and implementing a successful industrial policy. Although China adopted several elements from its neighbors, it did not follow the traditional path of other Asian states precisely, due to its particular administrative organization and socialist economic structure.

More specifically, Chinese policy is structured around a hybrid organization that combines elements of strong central guidance and administrative decentralization. Thus, while the central government holds tight fiscal control and sets general guidelines and policy strategies, there is significant local decentralization in the implementation and enforcement of these policies. Decentralization is based on a system of interregional competition, in which different local governments compete with one another based on their performance and effectiveness. The

performance criteria encompass both quantitative measures, such as increasing a specific percentage of GDP per region, and qualitative objectives, including ensuring stability. This structure, which functions more like a private enterprise (Meier, 2018), has led to a fragmented environment that deviates from the strict top-down logic of South Korea and Japan.

Beyond the administrative structure, hybridity is also evident in the economy. After 1978, the legacy of the centrally planned economy proved to be extremely dysfunctional. Thus, in the absence of appropriate conceptual tools for a smooth transition to a market economy, China followed a gradual process of opening up and reform. By maintaining the old system and slowly opening up the economy (dual track system), it allowed for experimentation and reduced the risks of a shock that could disrupt the country's social and political life (Naughton, 2007). In contrast to the pre-existing capitalist structures of South Korea and Japan, China had to rebuild institutional and conceptual tools to accommodate the new market economy. The result was a system in which socialist and capitalist elements coexist and interact. These two important hybridities shaped the direction of China's industrial policy both in the early period, before Xi's rise, and afterwards.

First Phase: Opportunistic Development and Security Issues.

With China's entry into the World Trade Organization in 2001, its development policy shifted its focus to managing the risks associated with integration into the globalized system. Thus, the initial strategy was based on opportunistic imitation policies aimed at catching up, without any clear coherence. Although this practice continued until at least 2013, GDP growth was impressive. It grew by 10% annually between 2003 and 2005, reaching 12.7% in 2006 and 14.2% in 2007. This growth was the result of the redistribution of productive resources (labor, capital, materials) as well as the free movement of internal migrants and their concentration in traditional industries.

This increase was the result of the redistribution of productive resources (labor, capital, materials) as well as the free movement of internal migrants and their concentration in traditional industrial sectors. In this context, in 2006, then-President Hu Jintao presented the first attempt at autonomous and domestic innovation (*zìzhǔ chuàngxī*, 自主创新) through the "National Medium- and Long-Term Plan for Science and Technology Development" (2006-2020). This 15-year program, to be implemented by local governments, provided guidance and funding for 16 major projects without specifying how they were to be executed. Its aim was to leverage existing technologies from advanced countries to develop traditional sectors while minimising costs and risks. Although it was not an organised industrial policy, as it lacked internal and unifying logic between sectors (Naughton, 2020), it did establish, for the first time, a top-down framework for a long-term strategy for innovation and technology transfer.

The 2008 economic crisis was a turning point for China's industrial and economic policies. It highlighted all the contradictions of the Chinese system, while laying the foundations for the structural changes that followed with Xi's rise to power in 2013. The crisis halted productivity, highlighted China's dependence on the West, and was framed as a failure of the American capitalist system. To address these challenges and stimulate domestic consumption, a four trillion yuan fiscal

stimulus package was proposed, with an emphasis on infrastructure investment. Thus, growth began to be fueled by consumption and real estate construction. However, the implementation strategy created several problems.

Central funds were primarily allocated to the maintenance of low-performing state-owned enterprises, and to a lesser extent, to local and private companies. At the same time, local governments, burdened with the task of implementing investments, had few resources at their disposal. China's Tax-Sharing system allocated a small percentage to local governments, while other institutional constraints, such as the inability to borrow from banks, limit the available financing tools. At the same time, the emphasis on GDP growth as the primary criterion for evaluating and advancing the careers of local officials increased the pressure for immediate action. Thus, in an attempt to circumvent economic and institutional constraints, they created entities—Local Government Financial Vehicles—which acted as intermediaries and financing tools for raising funds from banks.

This created a fragmented but flexible financial environment, structured around personal connections and local alliances. The result was the creation of multiple internal crises. On the one hand, there was the creation of excessive local and hidden debt, and on the other, overproduction in the steel, coal, and housing sectors, which did not meet actual demand. Scattered ghost towns in western regions and urban centers with incomplete and uneven development, as well as "urban villages," called into question the structure of China's internal security.

New organizational principle after Xi: Security-Quality-Technology

The opportunistic structure of development, successive crises, and trade wars that followed the 2008 economic crisis necessitated a transformation of the country's political model and industrial strategy. The terms of the transformation were shaped by two interdependent concepts: a new holistic understanding of security and the transition from quantitative to qualitative growth.

Specifically, the shift towards a holistic approach to security (zǒngtǐ guójiā ānquán guān, 总体国家安全观) was introduced by Xi in April 2014, with a focus on technology. Until 2013, China lacked a coherent strategy in the field of technology and had a limited presence in global cyberspace and cybersecurity policy-making. However, significant developments in machine learning growth in 2012, along with advances in voice recognition and computer vision (Naughton, 2020), highlighted the revolutionary potential of technology in social, economic, and productive structures. At the same time, Snowden's revelations about the US government's collection of personal data, combined with Microsoft's discontinuation of support for Windows XP (Creemers, 2021), urgently raised the issue of cyber-sovereignty (wǎngluò zhǔquán, 网络主权) and cybersecurity (wǎngluò ānquán, 网络安全) of critical infrastructure related to technological development and cyberspace.

The new approach to security was not limited to the traditional meaning of the term. Instead, it was defined as a holistic (zǒngtǐ, 总体) and long-term dynamic process, which continuously

integrates different areas of everyday and scientific interest. At the same time, it was established as a basic organizational principle for high-quality development (gāo zhiliàng fāzhǎn 高质量发展) for a better life (měi hào shēnghuó, 美好生活). The shift towards quality development represents a structural change from the previous rapid and opportunistic approach. The goal now is to create a sustainable, innovative, and efficient state, focusing on technology, green energy, and efficiency, to unlock internal growth. Quality development, security, and technology are recurring themes that permeate the entire spectrum of Chinese policy.

Under these structural changes, new conceptual tools, institutions, and laws have been created to achieve two basic goals: 1) technological independence and self-reliance in critical areas such as semiconductors, and 2) technological dominance in global hi-tech production. These directions reflect the entire formation of China's industrial policy from 2014 to the present.

Methods and Means

Under the conceptual direction of domestic technological self-reliance (kējì zìlǐ, zìqiáng, 科技自立自强) and independent innovation (zìzhǔ chuàngxīn, 自主创新), new regulations and institutions were established to define the new relationships between the state, private enterprises, and technology.

The establishment of the Cyberspace Administration of China in 2014, along with the Central Cyberspace Affairs Commission, chaired by Xi, and the implementation of the country's first cybersecurity law, were decisive moments in shaping domestic technology policy. During the same period, the Made in China 2025 and Internet Plus initiatives were presented as national strategies for achieving two important long-term goals: 1) implementing new technologies (Internet of Things, cloud computing, Big Data) and integrating industrial robots into traditional sectors (transportation, electronics) in order to modernize and automate all stages of the production process. 2) reducing dependence on foreign countries and developing domestic innovation in sectors critical to the survival of the Chinese economy, such as AI and semiconductors.

The above initiatives, initially presented as separate strategies, were later integrated into a unified, long-term master plan: the "Innovation-driven Development Strategy" (IDDS) (CCP Central Committee and State Council, 2016). This plan serves as a guideline for transforming the entire economy by 2050 and, by extension, forms the organizational foundation for all upcoming five-year plans. Its key points concern the creation of China as an innovative state by 2025, a pioneer by 2030, and a superpower by 2050. However, the strategies presented in both the IDDS and the 13th and 14th plans are quite general, with no specific method of implementation. Instead, a general guiding strategy is provided that encompasses multiple policy cycles, with an emphasis on decentralized experimentation and flexibility in response to the evolving geopolitical circumstances.

The emphasis on innovation is central to the sustainability of Chinese industrial policy. After 2008, growth was mainly driven by consumption and infrastructure investment, which revealed their

limits and constraints. With the introduction of innovation into traditional sectors and the adoption of the new model, the goal is continuous optimization to reduce production costs and dominate international markets, while avoiding the 'flying geese' phenomenon. This was clearly evident in the case of Shaodong, a city in Hunan province. There, systematic innovation in the production of lighters resulted in the production of 15 billion pieces each year, accounting for approximately 70% of global demand.

At the same time, during the 13th (2016-2020) and 14th five-year plans (2020-2025), new financial institutions were created, such as the "State Fund for Industrial Guidance," and particular emphasis was placed on supporting private companies, which were recognised as the driving force behind the national effort. However, innovation was not left solely to individual initiative. To achieve technological and industrial self-sufficiency, scientific centers, university institutions, and state and private enterprises were organized into "innovation consortia" (Chuàngxīn liánhé tǐ, 创新联合体). As emphasized in the five-year plans:

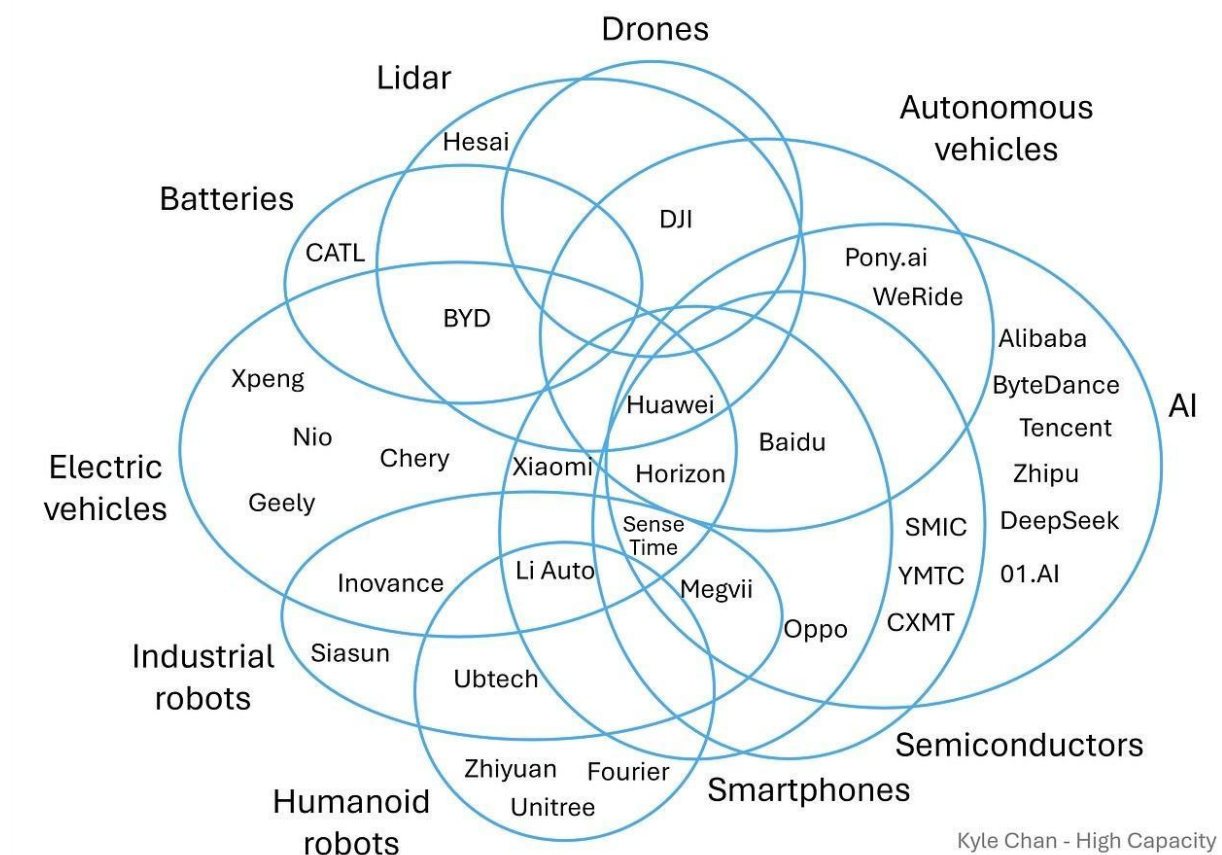
“research in certain interdisciplinary fields of study, and make use of existing advanced facilities to develop comprehensive national science centers. Enterprises, universities, and research institutes will be entrusted with building national technological innovation centers, and we will support the development of various entities involved in innovation, establishing an innovation network that integrates the efforts of the government, enterprises, universities, research institutes, and end-users. We will strengthen the position of enterprises as principal entities for innovation, as well as the leading role of enterprises in innovation, and encourage them to conduct basic and frontier research (13th Five-Year Plan, pp. 23-24)

Key general-purpose technology platforms facilitate the joint development of national industrial innovation centers by industry-leading enterprises in collaboration with institutions of higher education, scientific research institutes, and upstream and downstream industry partners, and undertake major national scientific and technological projects. “ (14th Five-Year Plan, p. 20)

Innovation consortia are scattered across various cities and regions in China and are typically task-oriented, focusing on the qualitative development of specific cutting-edge technologies. Under the framework of decentralized experimentation, they compete with each other for leadership in areas such as artificial intelligence, biopharmaceuticals, or new forms of energy. Success is equated with the application of innovation at the national level and the concentration of new resources in the most efficient regions. However, this interaction is not limited to their competitive relationship. These are not closed systems but structures that encourage cross-sectoral cooperation with other cities and regions. For example, the development of AI, EVs, and industrial robots is not seen as the evolution of individual sectors in different provinces, but rather as different parts of a whole that feed into each other to change the entire productive structure of the country. This is vital to Chinese industrial strategy, as it reduces overall timescales at all stages of the production process

and enhances effective cooperation and interdependence between different industries. This approach is not new to Chinese policy. It is based on the socialist concept of the "whole nation approach" (xinxing juguo tizhi, 新型 举国体制) and the widespread utilization of collective resources to achieve significant objectives.

Figure 2. Innovation Ecosystem: Interdependence of High Technology.



Source: Kyle Chan - High Capacity.

This hybrid relationship of competition and cooperation is subject to direct oversight by the state. As mentioned above, the state establishes the guiding framework through its financial instruments and identifies strategic areas that align with its political, industrial, and geopolitical objectives. At the same time, it allows the forces of competition and experimentation to bring about technological innovation in a protected environment. This structure is important for China, as it creates plasticity and flexibility in policy. The system has the ability to open and close, depending on economic and geopolitical conditions, and to regulate the limits of domestic consumption and exports.

Regulatory State: Shift towards domestic consumption

However, the presence of the state as a guiding institution is not limited to defining sectors and industries, nor is it exhausted in organizing internal competition. Its ability to regulate the direction of industrial policy in domestic or/and abroad, as well as strategic policies to handle overcapacity, also plays a crucial role. This dynamic became apparent in 2020 with the introduction of the 14th five-year plan. The experience of the first trade war in 2018, the coronavirus crisis in 2020, and the accompanying restrictions on exports changed the direction of industrial strategy towards greater introversion and systematization. The shift towards introversion was accompanied by the recognition that the weakness of Chinese domestic consumption is a critical security issue. In 2020, Xi stated in his speech "Important Issues Concerning China's Strategies for Medium- and Long-Term Economic and Social Development":

We must address our weaknesses. That is, in areas and sectors related to national security, we must build a domestic supply system that is independently controlled, secure, and reliable, so that self-circulation (自我循环) at critical moments and ensure the smooth operation of the economy in extreme situations'' (Speech by Xi Jinping, April 10, 2020)

Under these circumstances, the 14th Five-Year Plan prioritizes domestic consumption as the cornerstone of its industrial policy, while maintaining international trade (external circulation) as a complementary approach to domestic consumption.

We must sustain and enhance our superiority across the entire production chain in sectors such as high-speed rail, electric power equipment, new energy, and communications equipment, and improve industrial quality; and we must tighten international production chains' dependence on China, forming powerful countermeasures and deterrent capabilities based on artificially cutting off supply to foreigners. Second, we must make up for our shortcomings. That is, in sectors and segments related to national security, we must build a domestic supply system that is independently controllable and secure and reliable, so that self-circulation (自我循环) can be accomplished at critical moments, and ensure that the economy operates normally in extreme situations.'' (Xi Jinping speech, April 10, 2020)

By implementing dual circulation (domestic and international), the goal is to achieve self-sufficiency in the domestic industrial system at all stages of the production process, with a focus on domestic consumption. In areas where this is not possible, the external cycle, must fill this gap.

accelerate the establishment of a new development pattern (新发展格局) with domestic great circulation (国内大循环) as the mainstay and featuring mutually reinforcing domestic and international dual circulation (双循环); (14th Five-Year Plan EN, p. 10)

The emphasis on domestic consumption does not come out of nowhere. Combined with the impact of geopolitical crises on exports, internal problems make this issue a priority. The collapse of the real estate market in 2021 exhausted the capacity for growth in new infrastructure and the secondary demand for physical products such as electronic devices. At the same time, population decline and aging, as well as concerns about an impending economic stagnation (the middle-income trap), brought the issue of economic stability to the forefront. In the spring of 2025, the Chinese Communist Party magazine *Quishi* published excerpts from Xi Jinping's statements, in which he described weak demand as a threat to economic security: "Increasing domestic demand is not only about economic stability but also economic security—it is a strategic move, not a short-term solution".

The shift towards domestic consumption does not focus exclusively on physical products, but rather emphasizes a transition towards services, the qualitative upgrading of domestic structures, and is organized through a series of strategies and policies. Supply-side strategies, such as the ‘renewal urban development strategy’, aim to renew public infrastructure, rebuild degraded areas, and increase access to quality services in remote rural areas. At the same time, the integration of new technologies, such as AI and Big Data, into existing structures and services is supported. Alongside the qualitative upgrading of structures and services, efforts have been made in recent years to reduce the social and economic barriers faced by vulnerable groups. Emphasis has been placed on strategic reductions in inequality as a means of advancing the middle class, in order to "unlock latent growth" and thereby increase overall consumption of services, such as tourism (e.g., the ice and snow economy).

[At a glance](#)

China's strategy shows that it is not aiming for the development of individual sectors, but rather for inter-institutional and cross-sectoral interdependence with a view to restructuring the entire productive structure of the economy. The capabilities developed during the era of manufacturing and electrical goods production have been instrumental in driving the hi-tech development of services. Thus, the cooperation and interdependence of different sectors enhanced innovation efficiency, forming a holistic model that emphasizes quality and safety. At the same time, institutional support, decentralized experimentation, and investments in strategic companies, such as the German industrial robot and automation systems manufacturer Kuka, created channels of learning, integration, and redesign for domestic production. The results are evident in China's recent dominance in the overall production processes of rare earths, lithium, and industrial robots. Their competitive advantage is not limited to having the resources, but also to controlling over 60% of the entire production process and 90% of the refining process.

These successes, however, come with challenges that are familiar to the Chinese system. The emphasis on specific sectors and the creation of national priorities in the field of e-commerce services and electric cars have led to oversupply, reducing profits. Anti-involution efforts are under discussion, while strategies to boost domestic consumption are constantly being developed.

Despite the urgent recognition of the need to boost domestic consumption, structural reforms are not decisive enough to bring about real change. The 2013 reform program had already identified the important structural problems. These included the insufficiencies in the tax system, the hukou registration system, the institutional system of local incentives, the management of state-owned enterprises, etc. However, concerns over a sudden economic destabilization or burdening the existing middle class have led to slow structural reforms and a gradual approach on those key issues. All these matters have been at the heart of the 15th Five-Year Plan (2026-2030) since June 2024 and not much deviation is expected. Minor corrective changes are envisaged for the restructuring of priorities, such as efforts to control EV production as a method of managing oversupply, as well as moves to channel more of the oversupply abroad.

In order to understand those processes and the overall policy of China better, we need to engage in critical self-reflection. Asia, and China in particular, is still viewed through the distorting lens of "West versus East" or the even more simplistic dichotomy of "Western versus Asian values." Thus, it remains trapped in the position of the external "Other" in the global system, rather than being recognized as an active and formative part of it (Franceschini and Loubere, 2024). Only if we accept that China is not an exogenous, "exotic" power, but an integral part of global economic and institutional interdependence, will we be able to develop more sober, informed, and effective policy-making. This will allow us to judge more accurately, cooperate with greater self-awareness, and learn as quickly as possible.

Bibliography

CSET (2021, May) People's Republic of China 14th Five-Year Plan for National Economic and Social Development and Long-Range Objectives for 2035 [Original translation]. Center for Strategic and International Studies.

Creemers R.J.E.H. (2020), China's conception of cyber sovereignty: rhetoric and realization. In: Broeders D. & Berg B. van den (Eds.) *Governing Cyberspace: Behavior, Power, and Diplomacy. Digital Technologies and Global Politics* Lanham: Rowman & Littlefield. 107-142., Available at SSRN: <https://dx.doi.org/10.2139/ssrn.3532421>

Government of the People's Republic of China. (2015, May 19). Made in China 2025 [in Chinese]. http://www.gov.cn/zhengce/content/2015-05/19/content_9784.htm

Meier, N. (2009). China – The New Developmental State?: An Empirical Analysis of the Automotive Industry (NED-New edition). Peter Lang AG. <http://www.jstor.org/stable/j.ctv9hj98j>

Naughton, B. (2020). Industrial policy in China (1978–2020). CECHIMEX Working Paper Series. https://docs.dusselpeters.com/CECHIMEX/Naughton2021_Industrial_Policy_in_China_CECHIMEX.pdf

National Development and Reform Commission (NDRC), People's Republic of China. (2021). Outline of the 14th Five-Year Plan (2021–2025) [White paper]. <https://en.ndrc.gov.cn/policies/202203/P020220315511326748336.pdf>

People's Daily. (2020, August 13). Theory commentary on innovation-driven development under the 13th Five-Year Plan [in Chinese]. <http://theory.people.com.cn/n1/2020/0813/c40531-31820466.html>

United Nations Industrial Development Organization (UNIDO). (2022). The role of science, technology and innovation policies in the industrialization of developing countries: Lessons from East Asian countries. UNIDO.

Wu, F. (2014). The Chinese economy: Transitions and growth. Routledge. https://www.google.gr/books/edition/The_Chinese_Economy/BZv6AQAAQBAJ

Xi Jinping Thought Research Center for Social Sciences (XJTRC). (2021, August 9). China's 14th Five-Year Plan: Strengthening the domestic base to become a superpower. MERICS Commentary. <https://merics.org/en/comment/chinas-14th-five-year-plan-strengthening-domestic-base-become-superpower>

Zhejiang Institute for Economic and Social Research (ESSRA). (2023). Innovation-driven development and technological self-sufficiency: Comparative insights between the 13th and 14th Five-Year Plans [in Chinese]. <https://www.essra.org.cn/?news/1454>

Zhejiang Provincial Government. (2021, August 9). Outline of the 14th Five-Year Plan [Full text, in English]. https://www.fujian.gov.cn/english/news/202108/t20210809_5665713.htm

Xinhua News Agency. (2025, March 2). China deepens innovation reforms under the 15th Five-Year Plan framework [in Chinese].